# Information Package

- The new methodology for determining requirements for a data warehouse system is based on business dimensions.
- By using the new methodology  the measurements and the relevant dimensions that must be captured and kept in the data warehouse.
- This is known as an information package for the specific subject.
- Example : information package for analysing sales for a certain business.
  - The subject here is sales.
  - The measured facts  are shown in the bottom section of the package diagram.
    - actual sales, forecast sales, and bud-get sales.
  - The business dimensions are shown at the top of diagram as column headings.
    - time, location, product, and demographic age group.
- Each of these business dimensions contains a hierarchy or levels.
  - For example, the time dimension has the hierarchy going from year down to the level of individual day. ( quarter, month, and week).
- These levels or hierarchical components are shown in the information package diagram.

# Information Package

Information packages enables to :

- Define the common subject areas
- Design key business metrics
- Decide how data must be presented
- Determine how users will aggregate or roll up
- Decide the data quantity for user analysis or query
- Decide how data will be accessed

***Information Subject:*** Sales Analysis

**Dimensions**

Hierarchies

| Time Periods | Locations | Products | Age Groups | | |
|---|---|---|---|---|---|
| Year | Country | Class | Group 1 | | |
| | | | | | |
| | | | | | |
| | | | | | |

**Measured Facts**: Forecast Sales, Budget Sales, Actual Sales

**Figure 5-4** An information package.

## *Information Subject:* Automaker Sales

**Dimensions**

Hierarchies / Categories

| Time | Product | Payment Method | Customer Demo-graphics | Dealer | |
|---|---|---|---|---|---|
| Year | Model Name | Finance Type | Age | Dealer Name | |
| Quarter | Model Year | Term (Months) | Gender | City | |
| Month | Package Styling | Interest Rate | Income Range | State | |
| Date | Product Line | Agent | Marital Status | Single Brand Flag | |
| Day of Week | Product Category | | House-hold Size | Date First Operation | |
| Day of Month | Exterior Color | | Vehicles Owned | | |
| Season | Interior Color | | Home Value | | |
| Holiday Flag | First Year | | Own or Rent | | |

**Facts:** Actual Sale Price, MSRP Sale Price, Options Price, Full Price, Dealer Add-ons, Dealer Credits, Dealer Invoice, Down Payment, Proceeds, Finance

**Figure 5-5** Information package: automaker sales.

## *Information Subject:* Hotel Occupancy

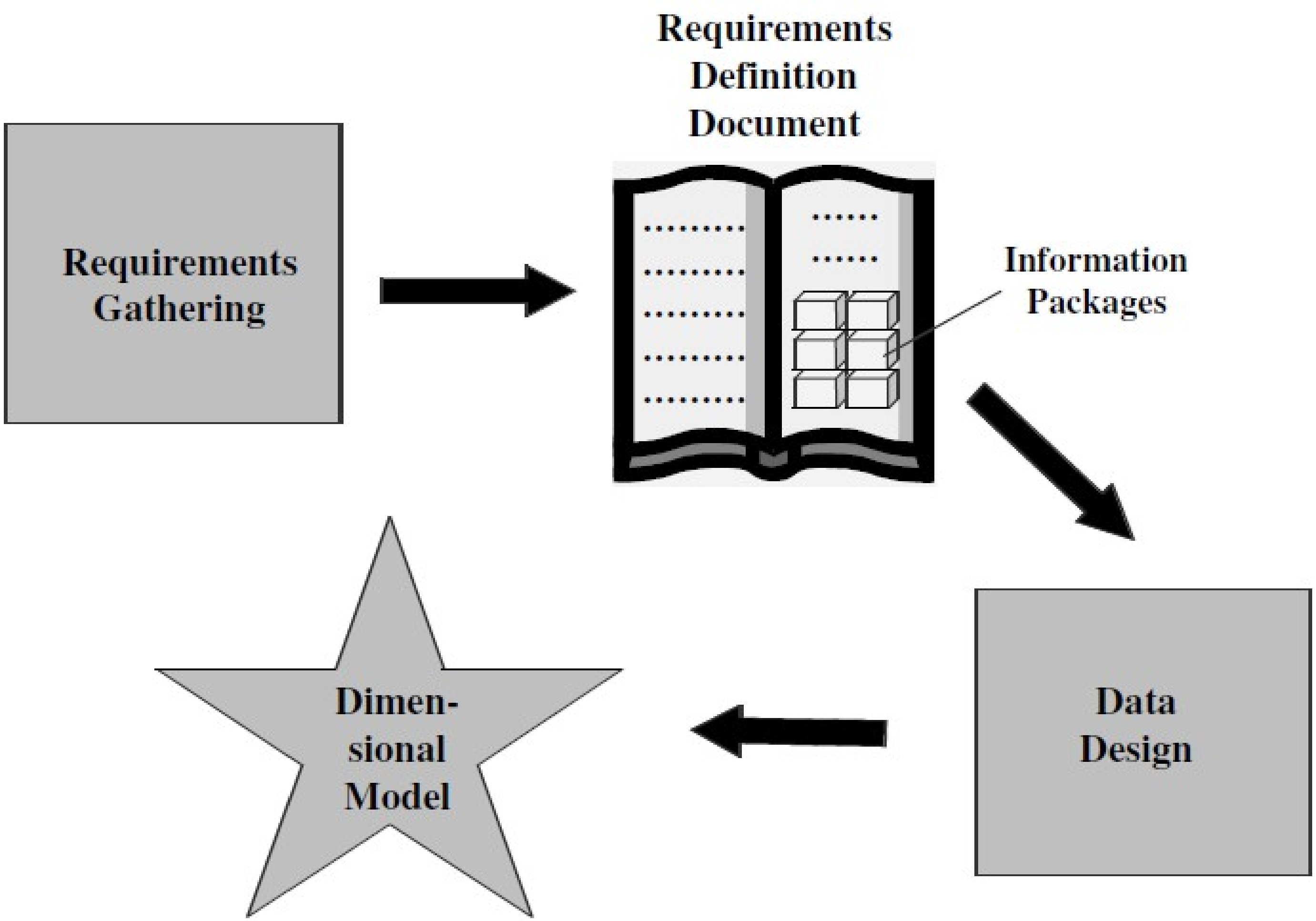


**Figure 10-1** From requirements to data design.

# DIMENSIONAL MODELING

- Design Decisions
- Choosing the process.
  - Selecting the subjects from the information packages for the first set of logical structures to be designed.
- Choosing the grain.
  - Determining the level of detail for the data in the data structures.
- Identifying and conforming the dimensions.
  - Choosing the business dimensions (such as product, market, time, etc.) to be included in the first set of structures.
- Choosing the facts.
  - Selecting the metrics or units of measurements (such as product sale units, dollar sales, dollar revenue, etc.) to be included in the first set of structures.
- Choosing the duration of the database.
  - Determining how far back in time you should go for historical data.

## Dimensions

**Automaker Sales**

**Fact Table**

Actual Sale Price
MSRP Sale Price
Options Price
Full Price
Dealer Add-ons
Dealer Credits
Dealer Invoice
Down Payment
Proceeds
Finance

| Time | Product | Payment Method | Customer Demo-graphics | Dealer | |
|---|---|---|---|---|---|
| Year | Model Name | Finance Type | Age | Dealer Name | |
| Quarter | Model Year | Term (Months) | Gender | City | |
| Month | Package Styling | Interest Rate | Income Range | State | |
| Date | Product Line | Agent | Marital Status | Single Brand Flag | |
| Day of Week | Product Category | | House-hold Size | Date First Operation | |
| Day of Month | Exterior Color | | Vehicles Owned | | |
| Season | Interior Color | | Home Value | | |
| Holiday Flag | First Year | | Own or Rent | | |

**Facts**: Actual Sale Price, MSRP Sale Price, Options Price, Full Price, Dealer Add-ons, Dealer Credits, Dealer Invoice, Down Payment, Proceeds, Finance

**Dimension Tables**

**Product**

Model Name
Model Year
Package Styling
Product Line
Product Category
Exterior Color
Interior Color
First Year

**Time**

Year
Quarter …………..

**Payment Method**

Finance Type
Term ……………....

**Customer Demo-graphics**

Age
Gender …………..

**Dealer**

Dealer Name
City …………..

| Time | Product | Payment Method | Customer Demo-graphics | Dealer | |
|---|---|---|---|---|---|
| Year | Model Name | Finance Type | Age | Dealer Name | |
| Quarter | Model Year | Term (Months) | Gender | City | |
| Month | Package Styling | Interest Rate | Income Range | State | |
| Date | Product Line | Agent | Marital Status | Single Brand Flag | |
| Day of Week | Product Category | | House-hold Size | Date First Operation | |
| Day of Month | Exterior Color | | Vehicles Owned | | |
| Season | Interior Color | | Home Value | | |
| Holiday Flag | First Year | | Own or Rent | | |

**Facts**: Actual Sale Price, MSRP Sale Price, Options Price, Full Price, Dealer Add-ons, Dealer Credits, Dealer Invoice, Down Payment, Proceeds, Finance

# Criteria for combining the tables into a dimensional model.

- The model should provide the best data access.
- The whole model must be query-centric.
- It must be optimized for queries and analyses.
- The model must show that the dimension tables interact with the fact table.
- It should also be structured in such a way that every dimension can interact equally with the fact table.
- The model should allow drilling down or rolling

# THE STAR SCHEMA

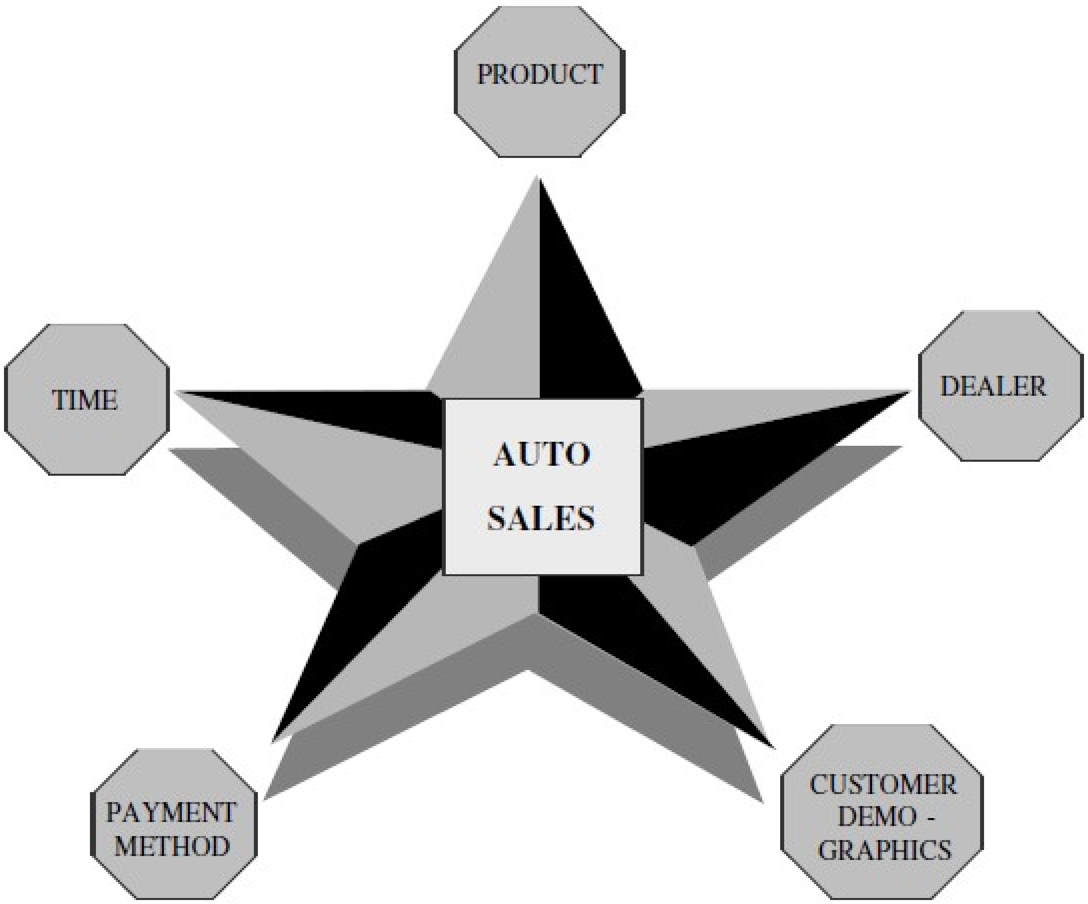


**Figure 10-4** STAR schema for automaker sales.

**Entity-Relationship Modeling**

Removes data redundancy
Ensures data consistency
Expresses microscopic relationships

**Figure 10-5** E-R modeling for OLTP systems.

**Dimensional Modeling**

Captures critical measures
Views along dimensions
Intuitive to business users

**Figure 10-6** Dimensional modeling for the data warehouse.

# E-R Modeling Versus Dimensional Modeling

E-R Modeling

- OLTP systems capture details of events or transactions
- OLTP systems focus on individual events
- An OLTP system is a window into micro-level transactions
- Picture at detail level necessary to run the business
- Suitable only for questions at transaction level
- Data consistency, non-redundancy, and efficient data storage critical

# E-R Modeling Versus Dimensional Modeling

Dimensional Modeling

- DW meant to answer questions on overall process
- DW focus is on how managers view the business
- DW reveals business trends
- Information is centered around a business process
- Answers show how the business measures the process
- The measures to be studied in many ways along several business dimensions

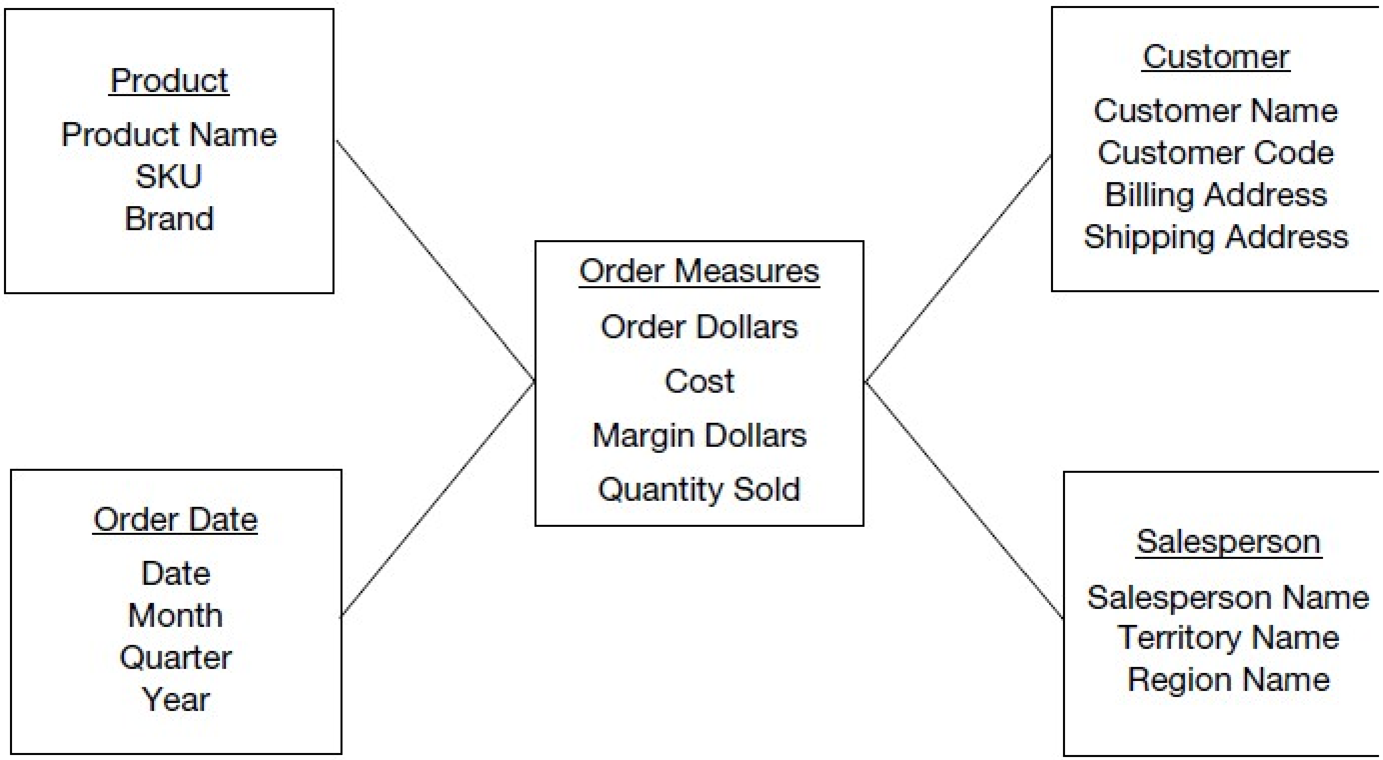


**Figure 10-7** Simple STAR schema for orders analysis.

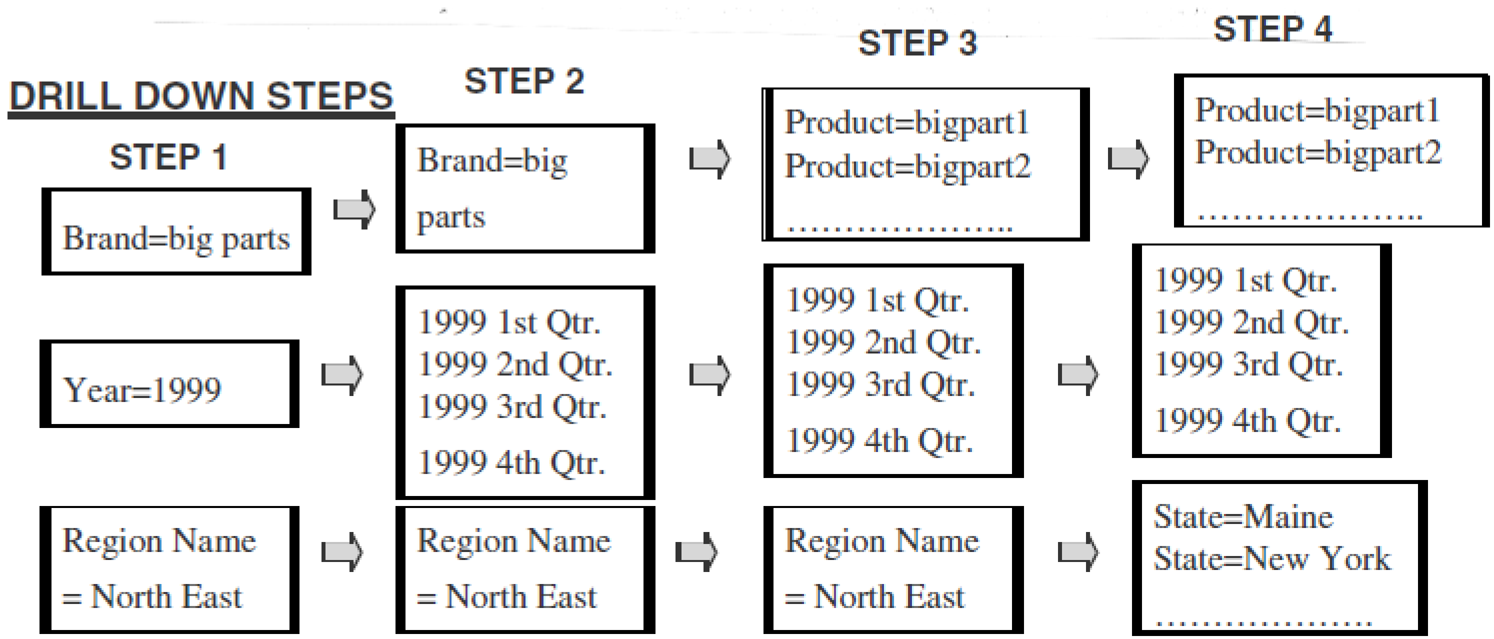


**Figure 10-9** Understanding drill-down analysis from the STAR schema.

# Inside a Dimension Table

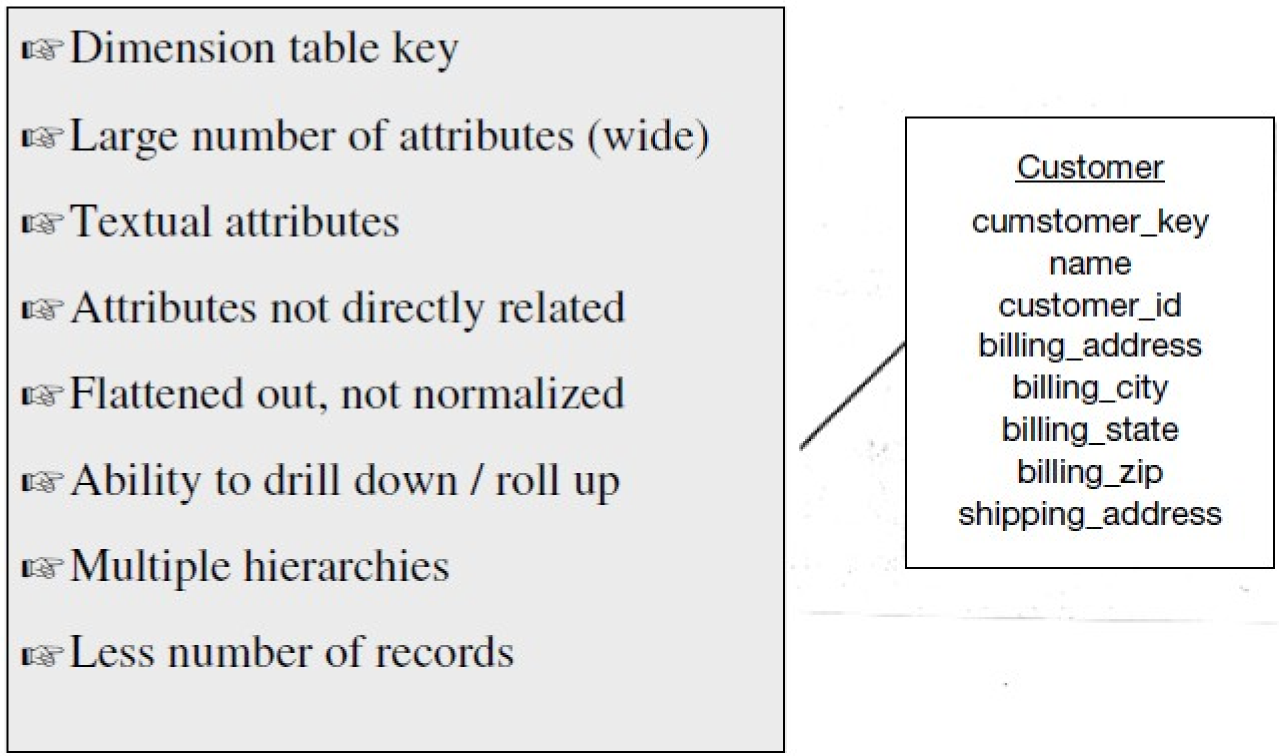


**Figure 10-10** Inside a dimension table.

# Inside the Fact Table

- Concatenated fact table key
- Grain or level of data identified
- Fully additive measures
- Semi-additive measures
- Large number of records
- Only a few attributes
- Sparsity of data
- Degenerate dimensions

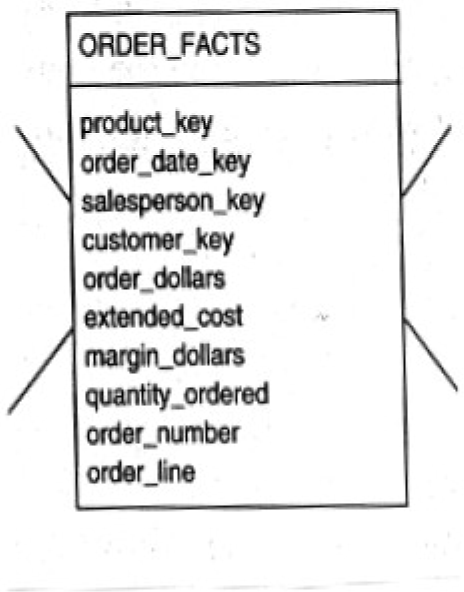


**Figure 10-11** Inside a fact table.

# The Factless Fact Table

- For analyzing student attendance, the possible dimensions are

  student, course, date, room, and professor.
- The attendance may be affected by any of these dimensions.
- When you want to mark the attendance relating to a particular course, date, room, and professor, what is the measurement you come up for recording the event?
- In the fact table row, the attendance will be indicated with the number *one*. Every fact table row will contain the number *one* as attendance.
- The very presence of a corresponding fact table row could indicate the attendance.
- This type of situation arises when the fact table represents events. Such fact tables really do not need to contain facts.
- They are “factless” fact tables.

# The Factless Fact Table

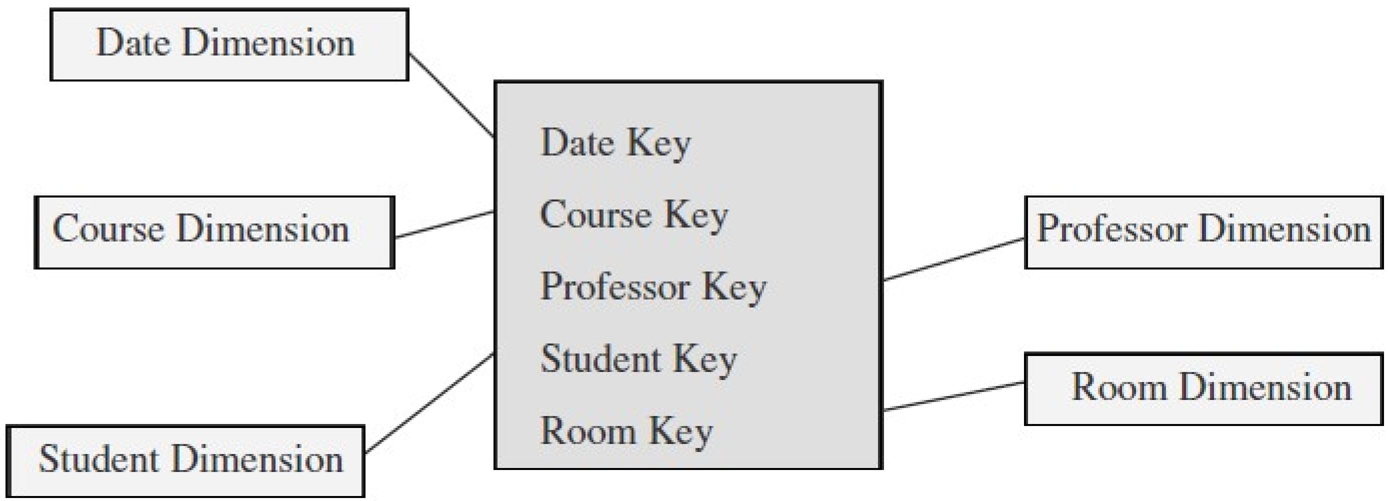


**Figure 10-12** Factless fact table.

# Keys in the Data Warehouse Schema

- **Primary Keys**
  - Each row in a dimension table is identified by a unique value of an attribute designated as the primary key of the dimension.
  - In a product dimension table-primary key identifies each product uniquely.
  - In the customer dimension table, the customer number identifies each customer uniquely.
  - In the sales representative dimension table, the social security number of the sales representative identifies each sales representative.

# Keys in the Data Warehouse Schema

- some implications of these candidate keys.
- Assume that the product code in the operational system is an 8-position code,
  - two of which positions indicate the code of the warehouse where the product is normally stored,
  - two other positions denote the product category.
- If we use the product code as the primary key.
  - If the product code gets changed in the middle of a year then
  - then the aggregate data before the change and after the change to the product code.

# Keys in the Data Warehouse Schema

- **Surrogate Keys**
  - The surrogate keys are simply system-generated sequence numbers.
  - They do not have any built-in meanings.
  - The surrogate keys will be mapped to the production system keys.
  - Keep the operational system keys as additional attributes in the dimension tables.

# Keys in the Data Warehouse Schema

# Keys in the Data Warehouse Schema

- **Foreign Keys**
- Each dimension table is in a one-to-many relationship with the central fact table.
- So the primary key of each dimension table must be a foreign key in the fact table.
- If there are four dimension tables of product, date, customer, and sales representative, then the primary key of each of these four tables must be present in the orders fact table as foreign keys.

# Keys in the Data Warehouse Schema

- The primary keys for the fact tables

1. *A single compound primary key whose length is the total length of the keys of the individual dimension tables.*
    - increases the size of the fact table (compound primary key, + the foreign keys).
2. *Concatenated primary key that is the concatenation of all the primary keys of the dimension tables.*
3. *A generated primary key independent of the keys of the dimension tables.*
    - Increases the size of the fact table (generated primary key)

# THE SNOWFLAKE SCHEMA

- “Snowflaking” is a method of normalizing the dimension tables in a STAR schema.
  - Partially normalize only a few dimension tables, leaving the others intact.
  - Partially or fully normalize only a few dimension tables, leaving the rest intact.
  - Partially normalize every dimension table.
  - Fully normalize every dimension table

# THE STAR SCHEMA

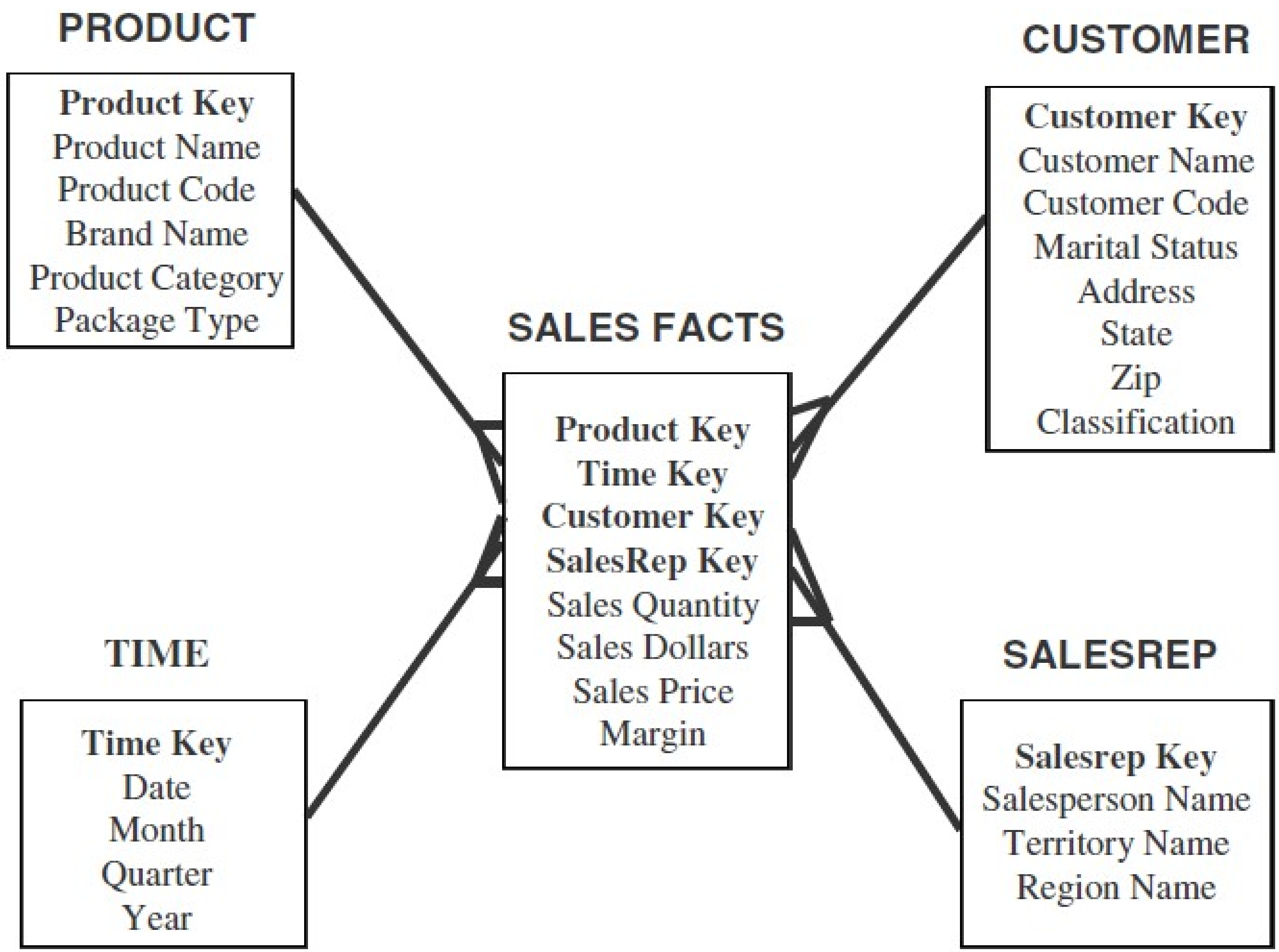


**Figure 11-7** Sales: a simple STAR schema.

# THE SNOWFLAKE SCHEMA

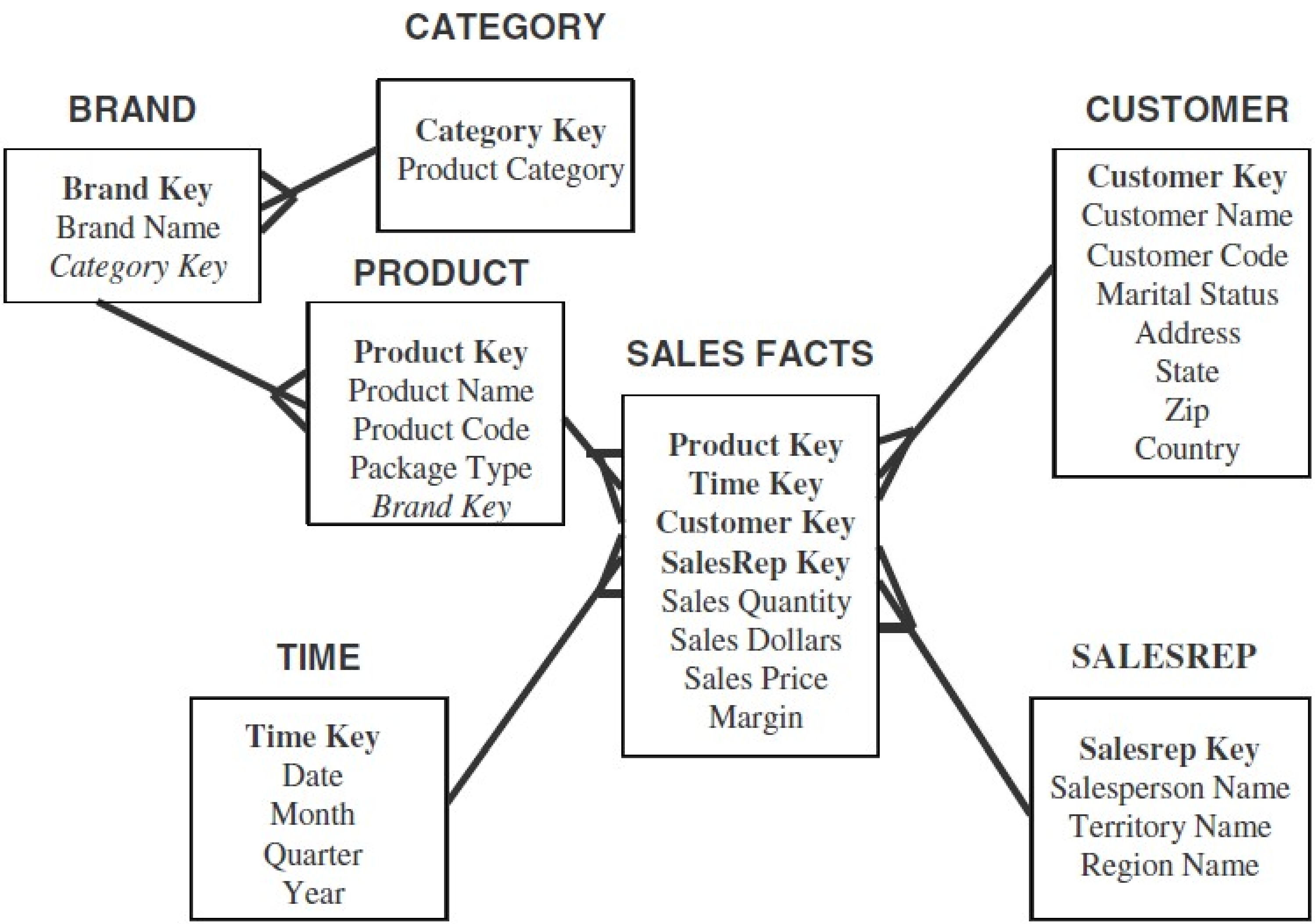


**Figure 11-8** Product dimension: partially normalized.

# THE SNOWFLAKE SCHEMA

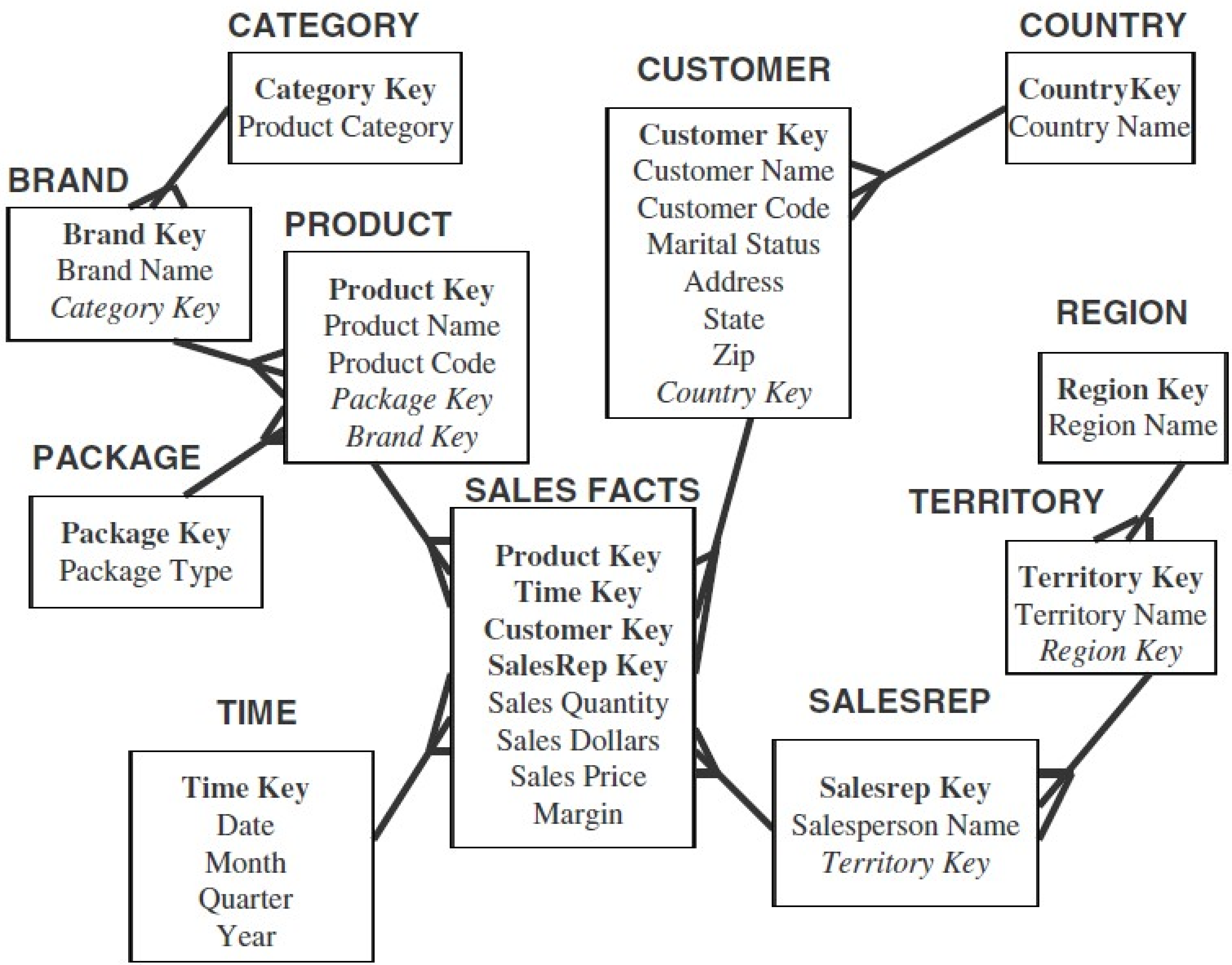


**Figure 11-9** Sales: “snowflake” schema.

# UPDATES TO THE DIMENSION TABLES

- The dimension tables are more stable and less volatile.
- The fact table, which changes through the increase in the number of rows
- A dimension table does not change just through the increase in the number of rows, but also through changes to the attributes themselves.
- If a particular product is moved to a different product category, then the corresponding values must be changed in the product dimension table.
- The types of changes that affect dimension tables and discuss the ways for dealing with these types.

# Slowly Changing Dimensions

- Most dimensions are generally constant over time
- Many dimensions, though not constant over time, change slowly
- The product key of the source record does not change
- The description and other attributes change slowly over time
- In the source OLTP systems, the new values overwrite the old ones
- Overwriting of dimension table attributes is not always the appropriate option in a
- data warehouse
- The ways changes are made to the dimension tables depend on the types of changes and what information must be preserved in the data warehouse

# Three types of dimension table changes

- Type 1 Changes: Correction of Errors
- Type 2 Changes: Preservation of History
- Type 3 Changes: Tentative Soft Revisions

# Type 1 Changes

- The general principles for Type 1 changes:
    - Usually, the changes relate to correction of errors in source systems
    - Sometimes the change in the source system has no significance
    - The old value in the source system needs to be discarded
    - The change in the source system need not be preserved in the data warehouse
- Method for applying Type 1 changes is:
  - Overwrite the attribute value in the dimension table row with the new value
  - The old value of the attribute is not preserved
  - No other changes are made in the dimension table row
  - The key of this dimension table or any other key values are not affected
  - This type is easiest to implement

# Type 1 Changes

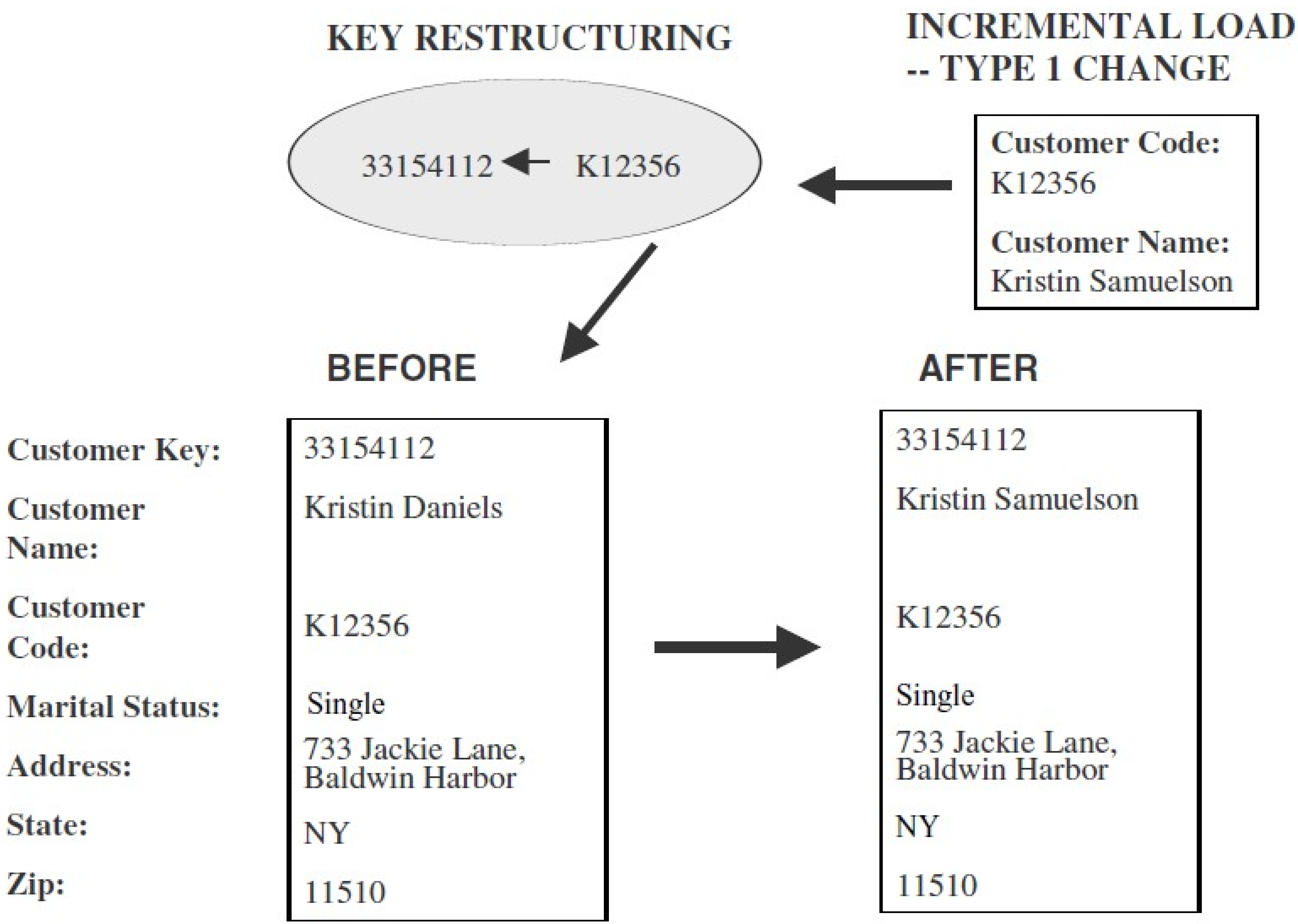


**Figure 11-2** The method for applying Type 1 changes.

# Type 2 Changes

- The general principles for this Type 2 of changes
  - They usually relate to true changes in source systems
  - There is a need to preserve history in the data warehouse
  - This type of change partitions the history in the data warehouse
  - Every change for the same attribute must be preserved
- The method for applying Type 2 changes is:
  - Add a new dimension table row with the new value of the changed attribute
  - An effective date field may be included in the dimension table
  - There are no changes to the original row in the dimension table
  - The key of the original row is not affected
  - The new row is inserted with a new surrogate key

# Type 2 Changes

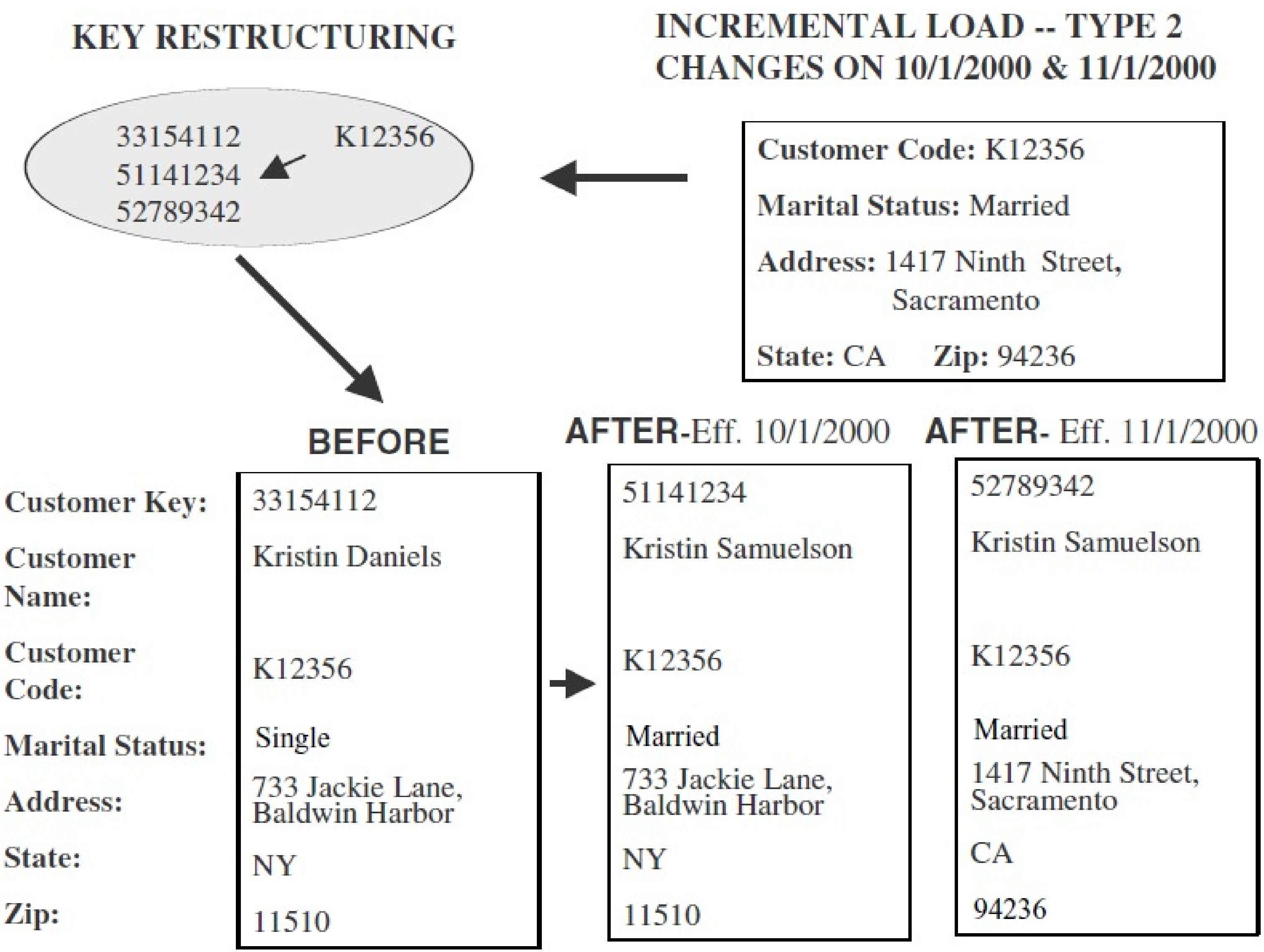


**Figure 11-3** The method for applying Type 2 changes.

# Type 3 Changes

- The general principles for Type 3 changes:
  - They usually relate to “soft” or tentative changes in the source systems
  - There is a need to keep track of history with old and new values of the changed attribute
  - They are used to compare performances across the transition
  - They provide the ability to track forward and backward
- The methods for applying Type 3 changes are:
  - Add an “old” field in the dimension table for the affected attribute
  - Push down the existing value of the attribute from the “current” field to the “old” field
  - Keep the new value of the attribute in the “current” field
  - Also, you may add a “current” effective date field for the attribute

# Type 3 Changes

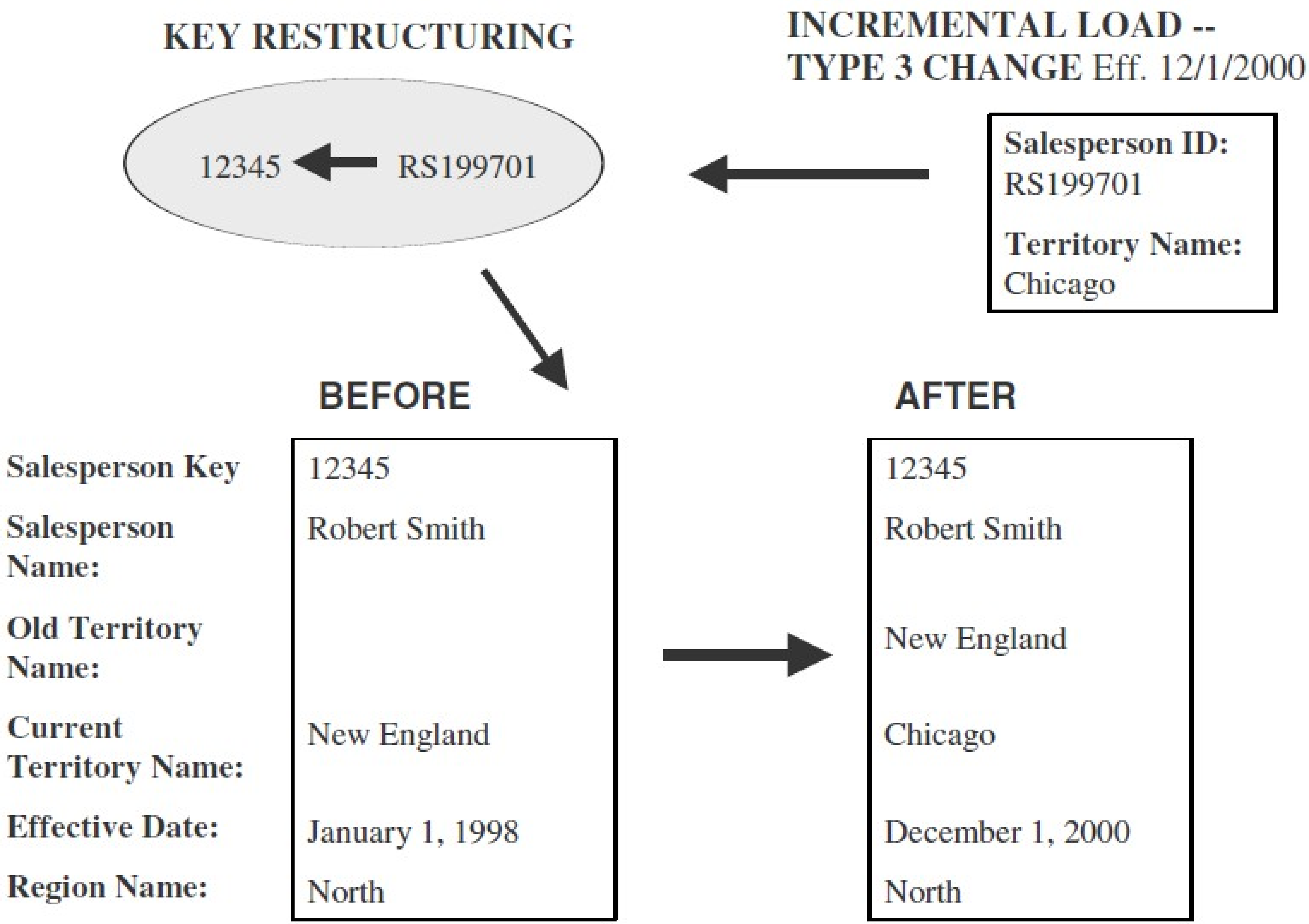


**Figure 11-4** Applying Type 3 changes.

# MISCELLANEOUS DIMENSIONS

- **Large Dimensions**
  - A large dimension is very deep(very large number of rows).
  - A large dimension may also be very wide ( large number of attributes).
  - Issues
    - Population of very large dimension tables
    - Browse performance of unconstrained dimensions, especially where the cardinality of the attributes is low
    - Browsing time for cross-constrained values of the dimension attributes
    - Inefficiencies in fact table queries when large dimensions need to be used
    - Additional rows created to handle Type 2 slowly changing dimensions

# Multiple Hierarchies

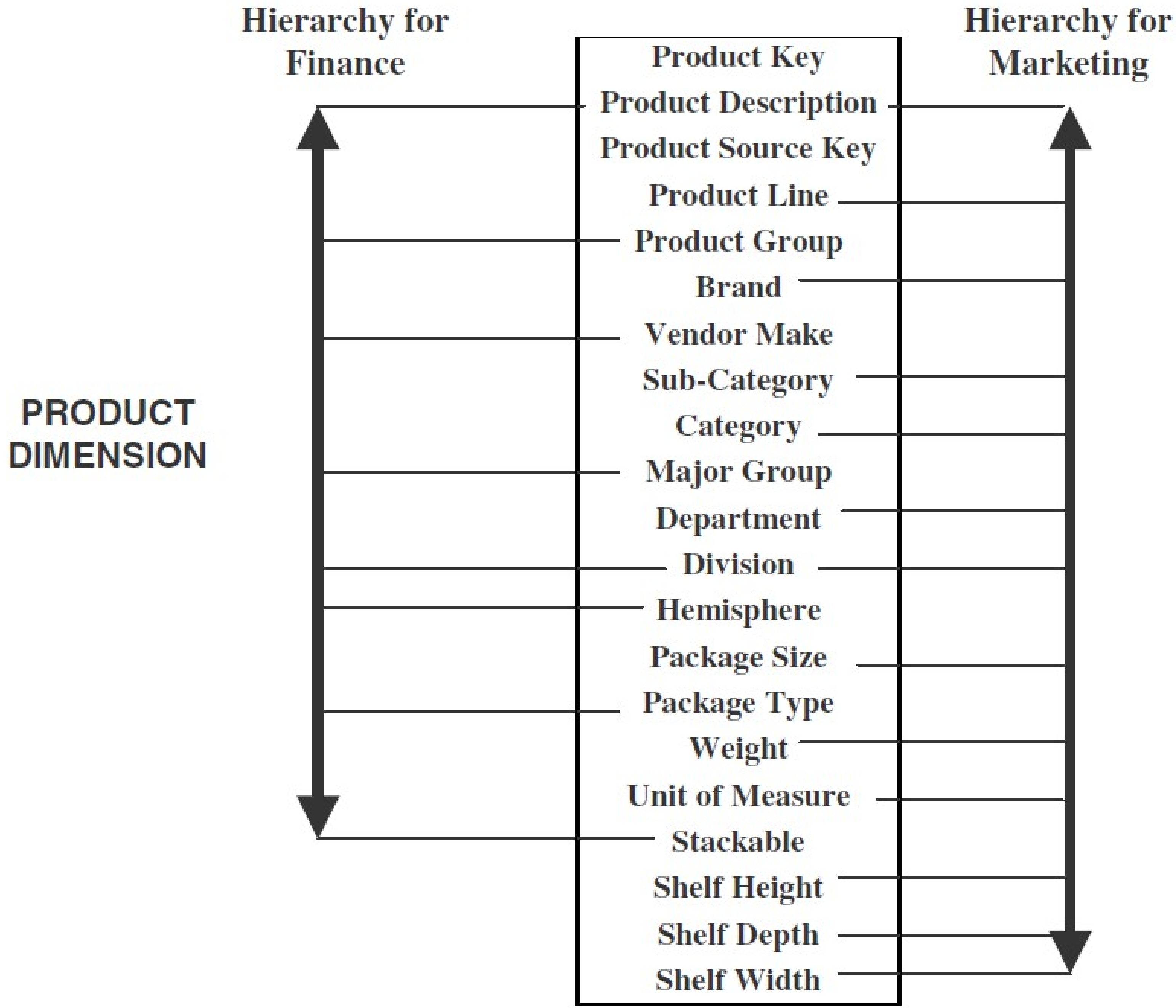


**Figure 11-5** Multiple hierarchies in a large product dimension.

# Rapidly Changing Dimensions

- Break off the rapidly changing attributes into another dimension table
- Leaving the slowly changing attributes behind in the original table.

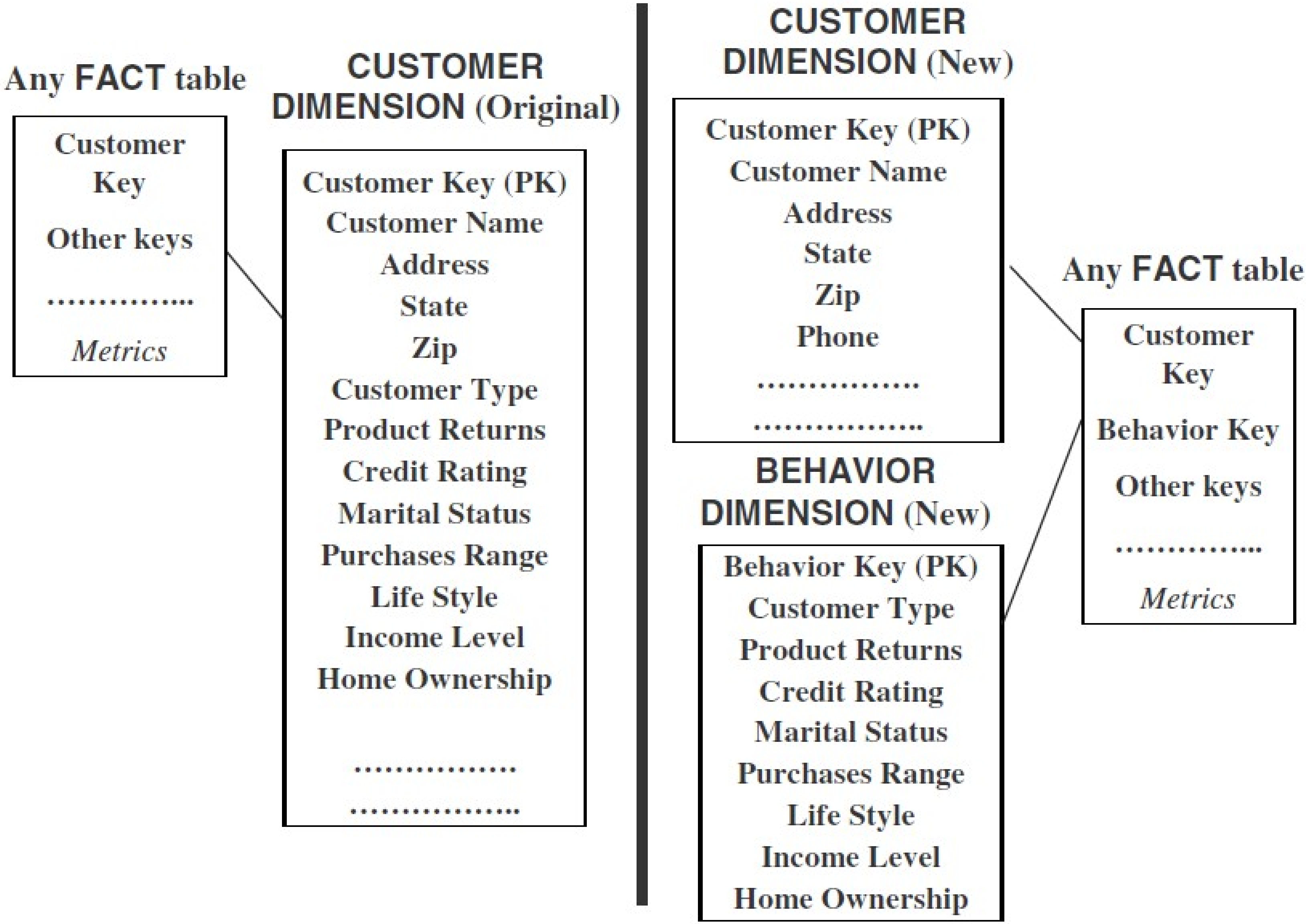


**Figure 11-6** Dividing a large, rapidly changing dimension table.

# Junk Dimensions

- Exclude and discard all flags and texts.
  - not a good option because you are likely to throw away some useful information.
- Place the flags and texts unchanged in the fact table.
  - This option is likely to swell up the fact table to no specific advantage.
- Make each flag and text a separate dimension table on its own.
  - The number of dimension tables will greatly increase.
- Keep only those flags and texts that are meaningful;
  - Group all the useful flags into a single "junk" dimension.
  - "Junk" dimension attributes are useful for constraining queries based on flag/text values

# AGGREGATE FACT TABLES

- Aggregates are precalculated summaries derived from the most granular fact table.
- These summaries form a set of separate aggregate fact tables.
- Create  each aggregate fact table as a specific summarization across any number of dimensions.
- Query : Total sales for all customers in the South-Central territory for the first two quarters of 2000 for product category Bigtools.
- Qualifying rows:
- All fact table rows where the customer key relates to all customers in the South-Central territory, the product key relates to all products in the product category Bigtools, and the time key relates to about 180 days in the first two quarters of 2000. A large number of fact table rows participate in the summation.

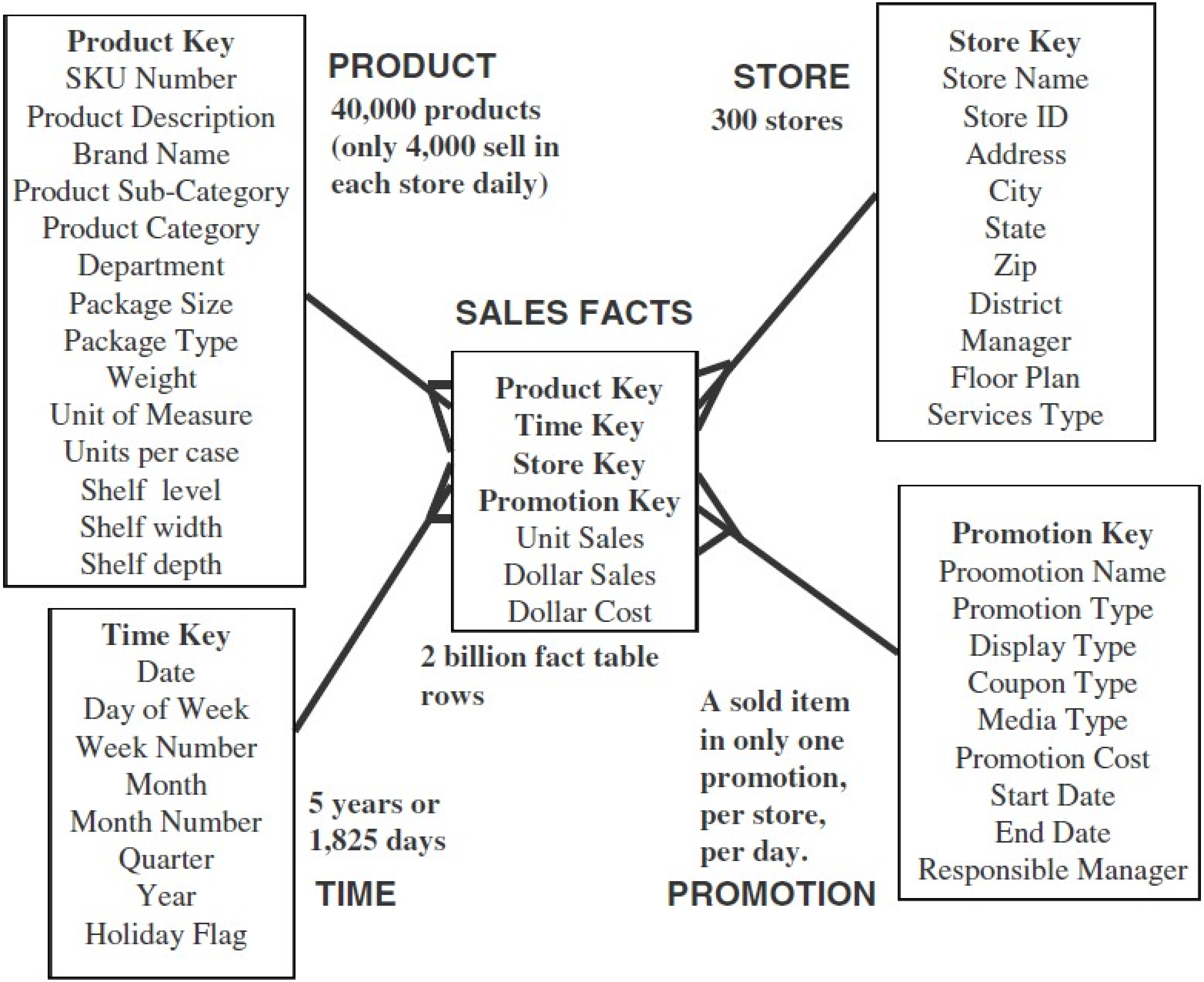


**Figure 11-12** STAR schema: grocery chain.

# *Multi-Way Aggregate Fact Tables.*

- *One-Way Aggregates.*
  - When you rise to higher levels in the hierarchy of one dimension and keep the level at the lowest in the other dimensions, you create one-way aggregate tables.
- *Two-Way Aggregates.*
  - When you rise to higher levels in the hierarchies of two dimensions and keep the level at the lowest in the other dimension, you create two-way aggregate tables.
- *Three-Way Aggregates.*
  - When you rise to higher levels in the hierarchies of all the three dimensions, you create three-way aggregate tables.

# Multi-Way Aggregate

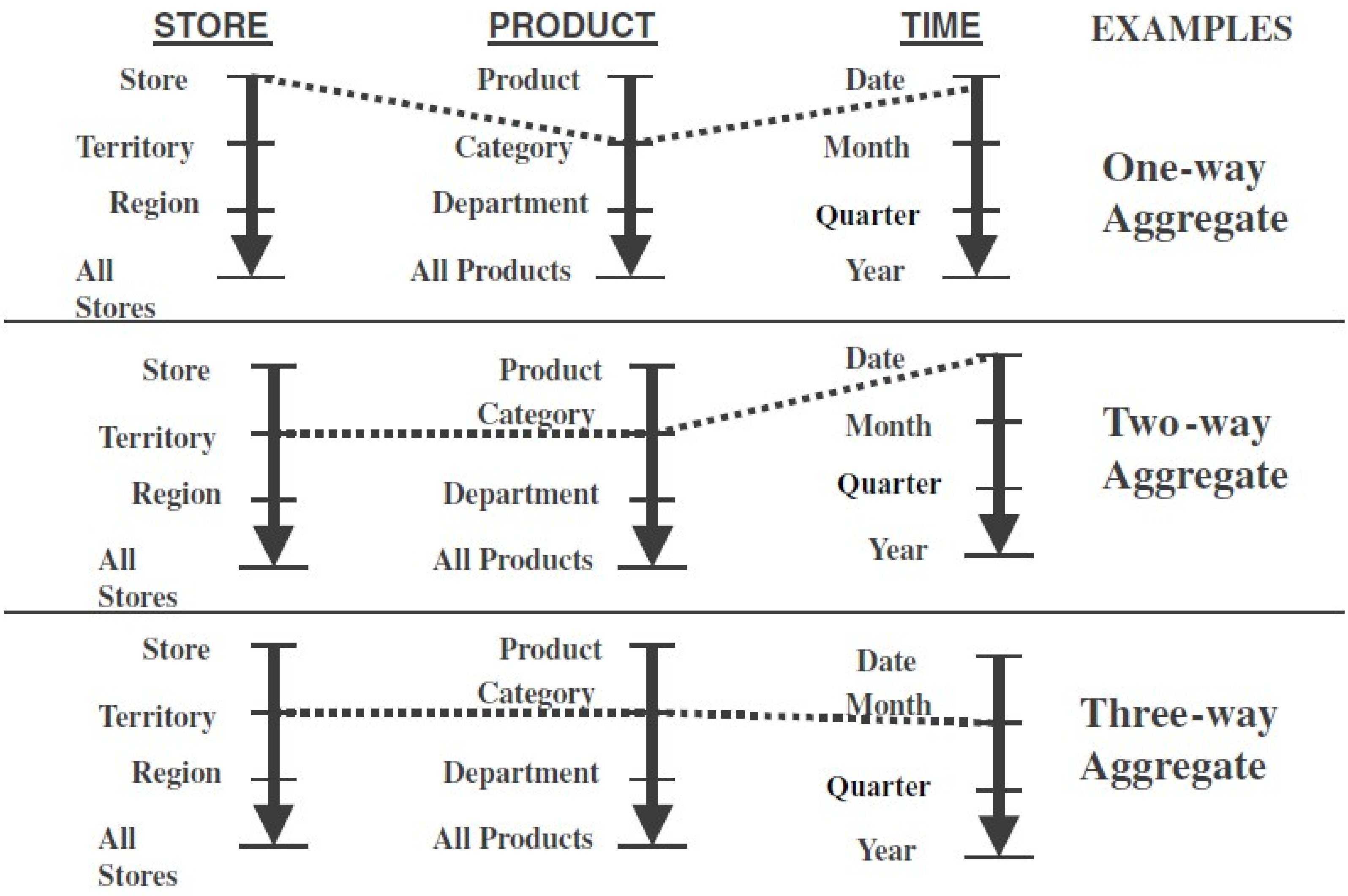


**Figure 11-14** Forming aggregate fact tables.

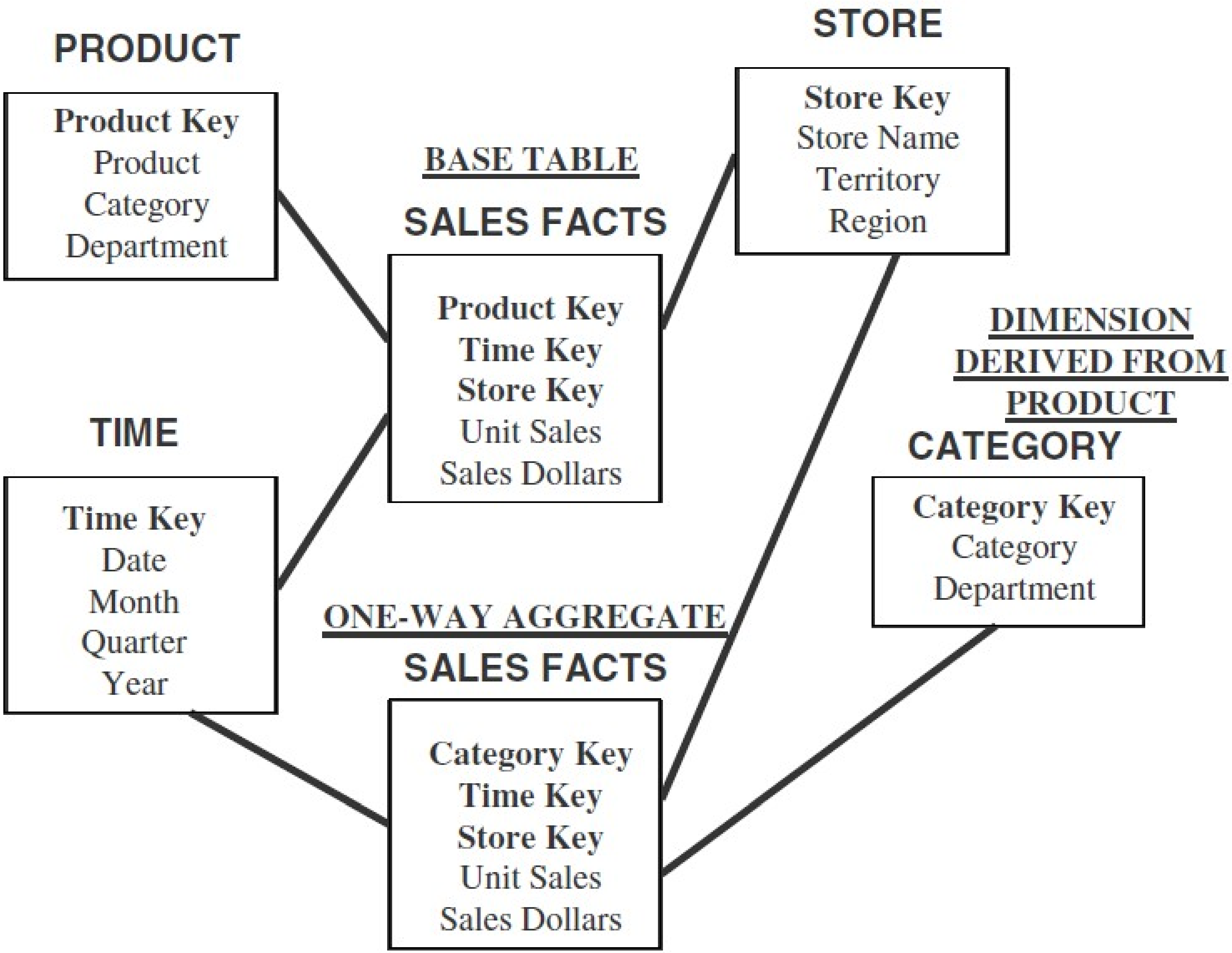


**Figure 11-15** Aggregate fact table and derived dimension table.

# Families Of STARS / Fact Constellation Schema

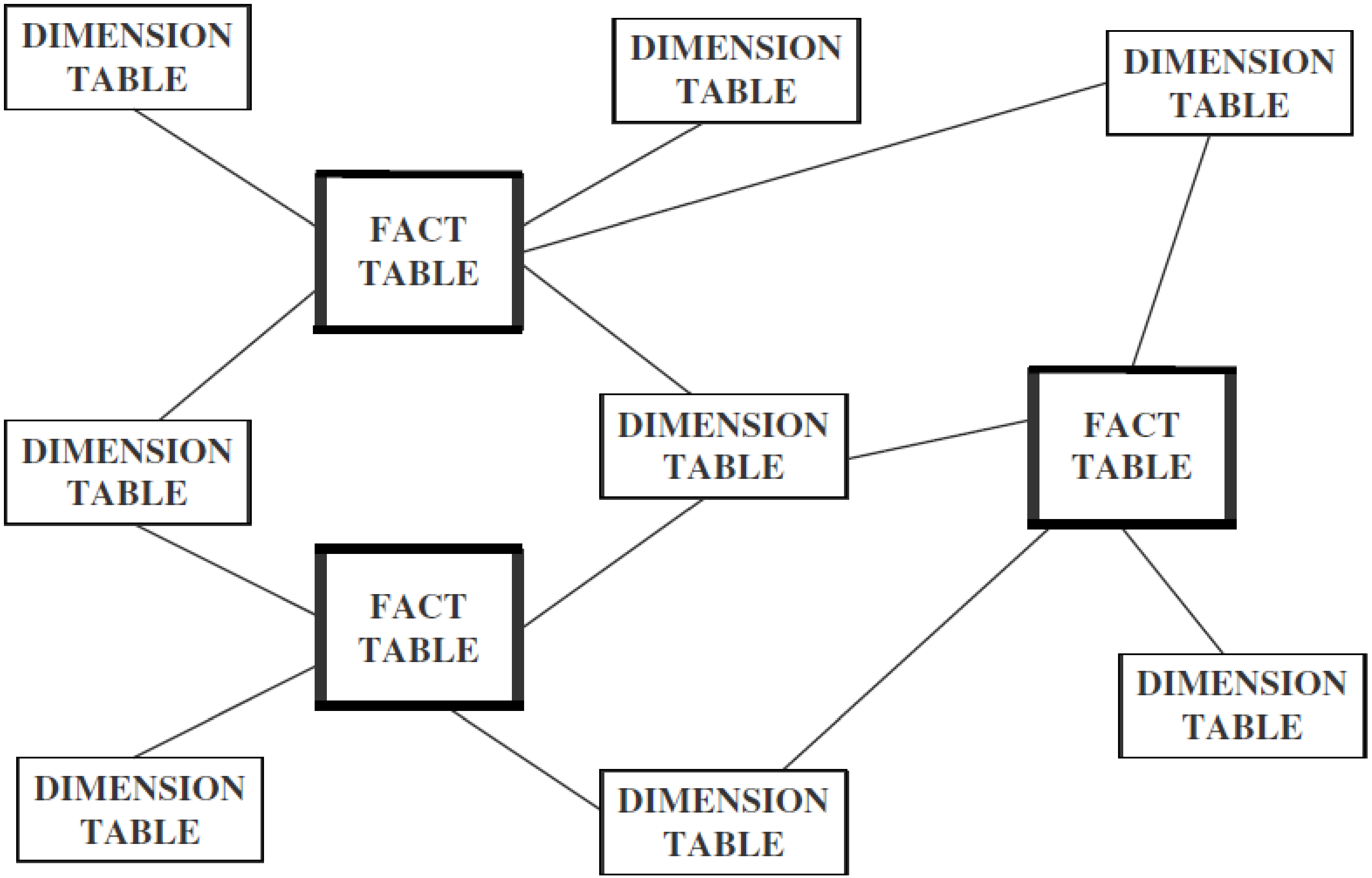


**Figure 11-16** Family of STARS.

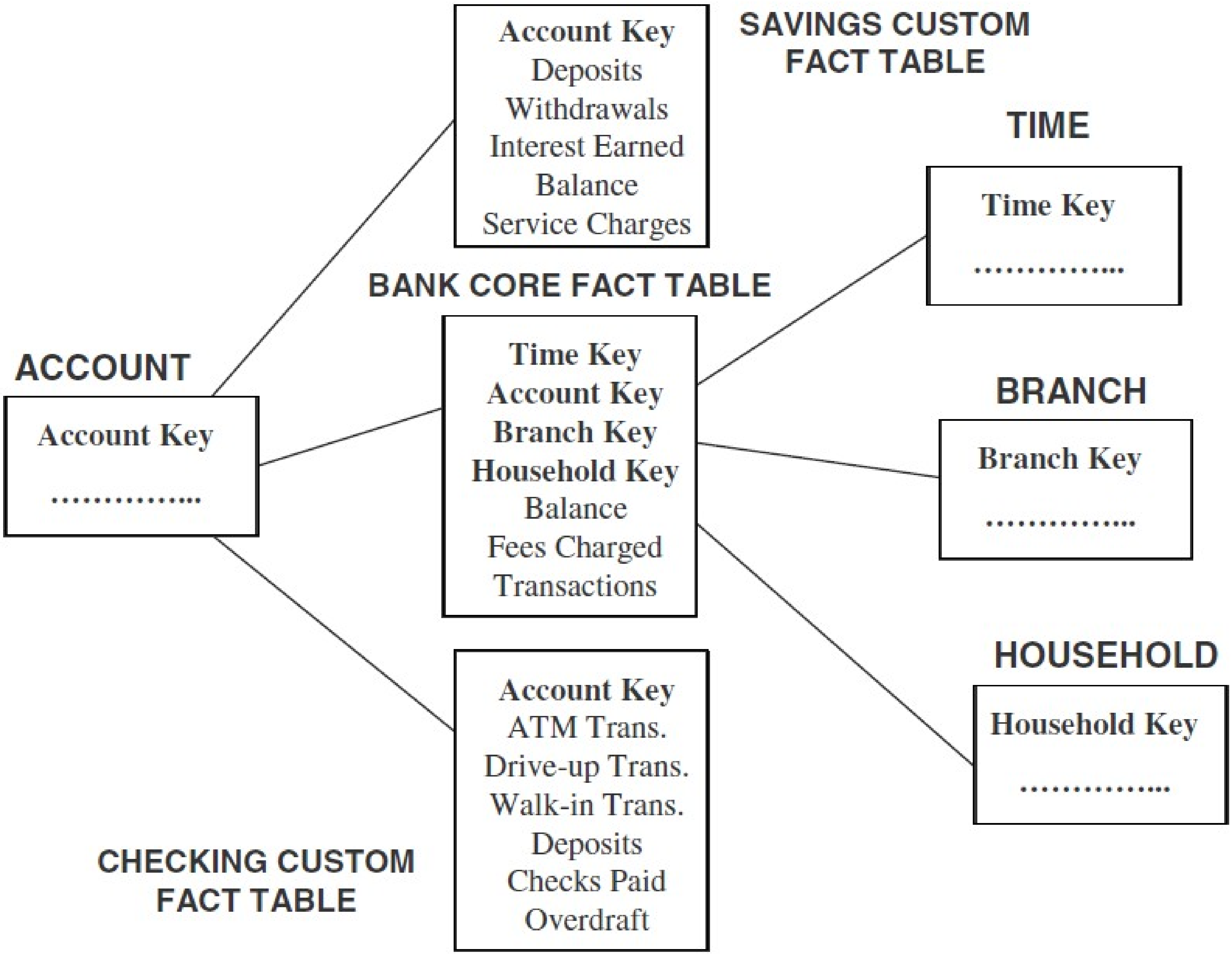


**Figure 11-18** Core and custom tables.

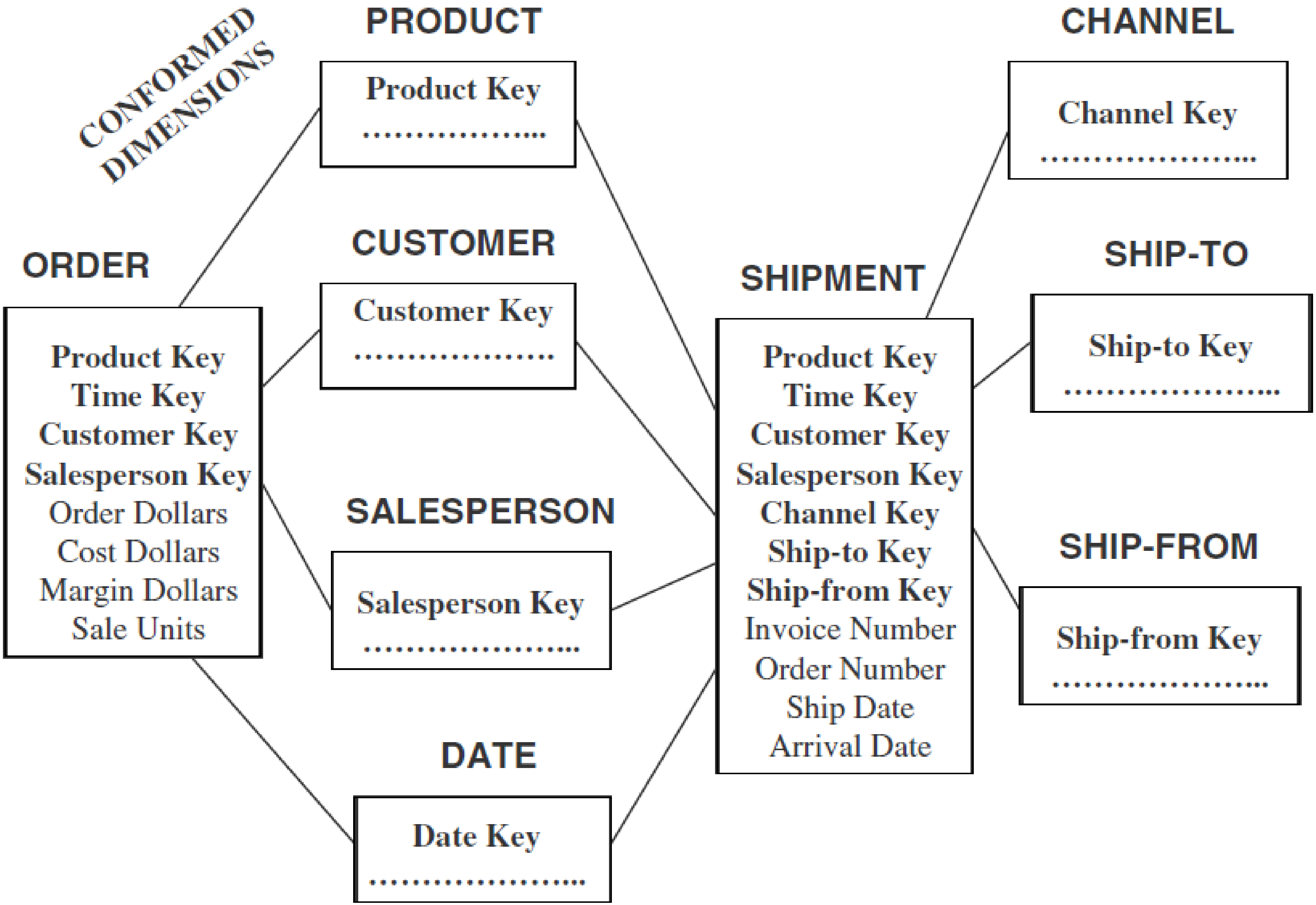


**Figure 11-19** Conformed dimensions.